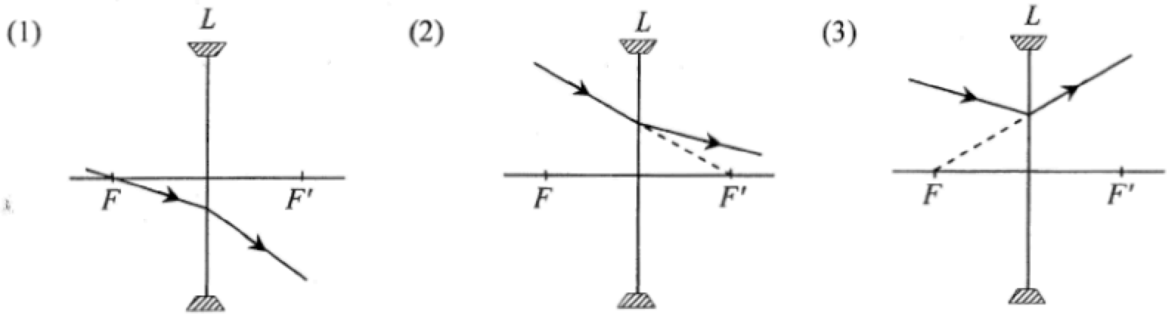
ch19 Lenses

2012 Q21

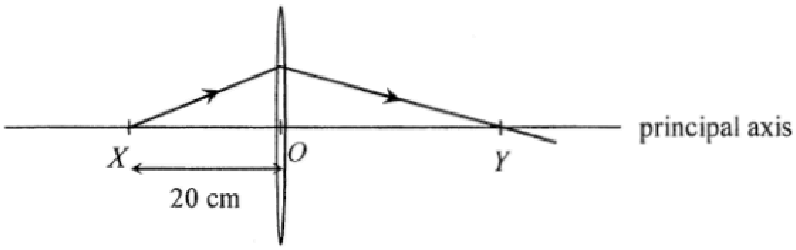
21. In each of the following diagrams, L is a concave lens and its two principal foci are denoted by F and F'. Which of the ray diagrams is/are possible ?



A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

2013 Q22

22.

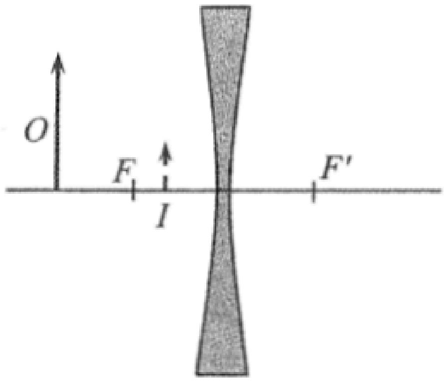


A point light source at X on the principal axis of a thin convex lens emits a ray of light. The ray passes through the lens and reaches the principal axis at point Y as shown. O is the optical centre of the lens such that $OX = 20$ cm and $OY > OX$. Which of the following statements is/are correct ?

(1) The focal length of the lens is shorter than 20 cm.
(2) If the point light source is shifted away from the lens, separation OY would increase.
(3) An object placed at Y would give a diminished image at X.

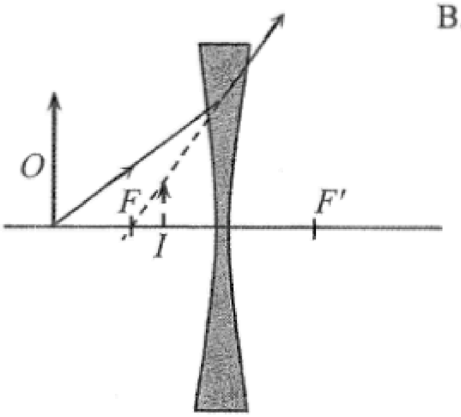
A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

15.

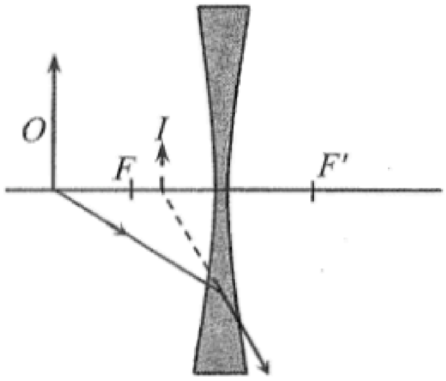


An object O placed in front of a concave lens forms an image I as shown. F and F' are lens. Which ray diagram is correct ?

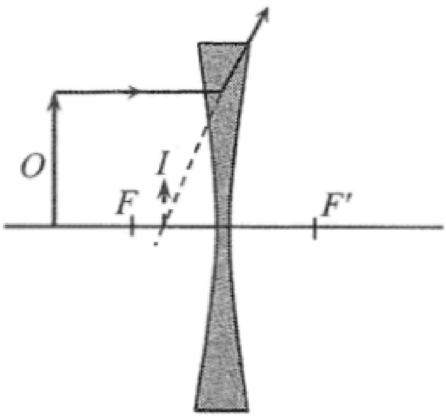
A.



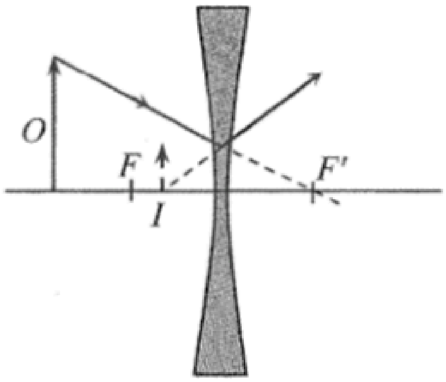
B.



C.

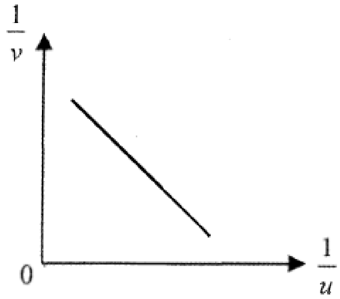


D.



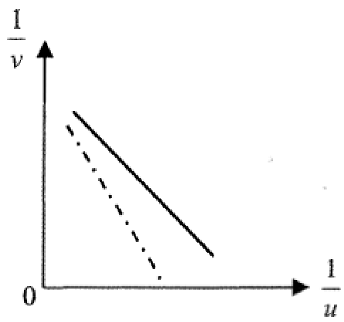
2015 Q16

*16.

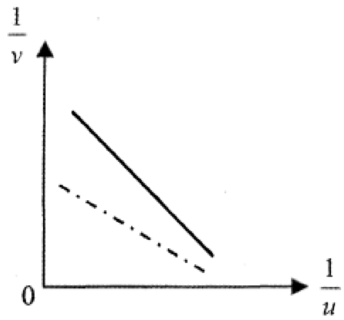


A student uses a convex lens to investigate the variation of image distance v with object distance u for real images. The graph of $\frac{1}{v}$ plotted against $\frac{1}{u}$ is shown above. If a convex lens of longer focal length is used, what would be the expected result (in dotted lines) ?

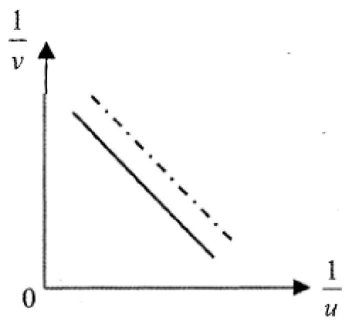
A.



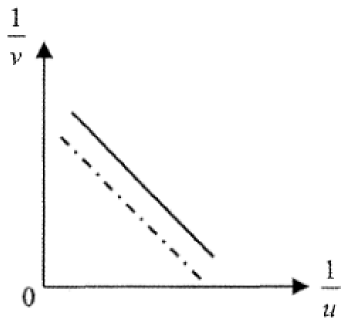
B.



C.



D.



2016 Q22

22. An object is moving at constant speed towards a convex lens of focal length 10 cm. At the moment when it is at 100 cm from the lens, which of the following descriptions of the image is correct ?

	direction of image movement	**speed of the image**
A.	away from the lens	faster than that of the object
B.	towards the lens	faster than that of the object
C.	away from the lens	slower than that of the object
D.	towards the lens	slower than that of the object

2017 Q19

*19. When an object is placed 30 cm in front of a concave lens, an image is formed 20 cm away from the lens. If the concave lens is replaced by a convex lens of the same focal length and the object distance remains unchanged, which of the following descriptions about the image formed is correct ?

	nature of the image	**image distance**
A.	real	20 cm
B.	real	60 cm
C.	virtual	20 cm
D.	virtual	60 cm

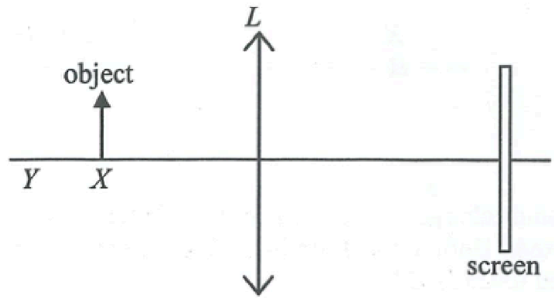
2018 Q19

*19. An object placed 25.0 cm in front of a lens forms a virtual image at a distance 11.1 cm from the lens. The lens is a

A. concave lens of focal length 7.7 cm.
B. concave lens of focal length 20 cm.
C. convex lens of focal length 7.7 cm.
D. convex lens of focal length 20 cm.

2019 Q20

20.

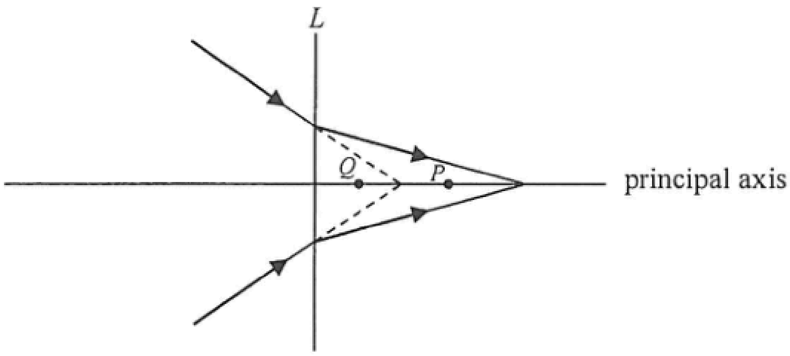


An object is placed at point *X* in front of a convex lens *L* as shown. A sharp image is captured by the screen. The object is then shifted to point *Y*. Which adjustment below may give a sharp image on the screen again ?

A. replacing *L* with another convex lens of longer focal length
B. replacing *L* with another convex lens of the same shape but made from material of a larger refractive index
C. replacing *L* with a concave lens
D. moving the screen to the right

2020 Q18

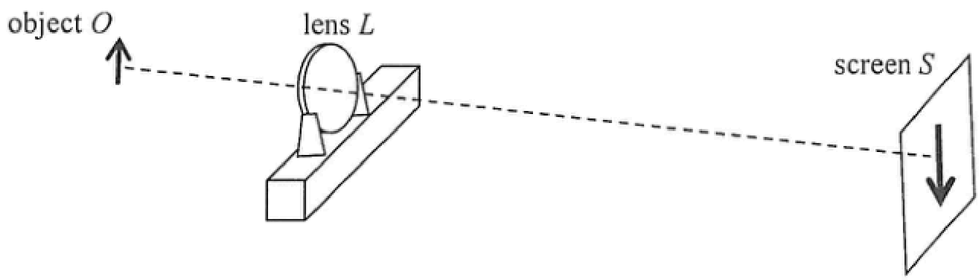
18.



Referring to the above ray diagram, what kind of lens is represented by *L* ? Which point, *P* or *Q* be its focus ?

	lens *L*	focus
A.	concave	*P*
B.	convex	*P*
C.	concave	*Q*
D.	convex	*Q*

20. The figure shows an enlarged sharp image of an object O formed on a screen S by a convex lens



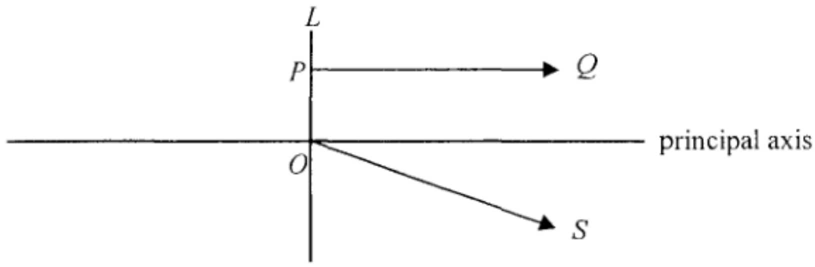
Which of the following can give a diminished sharp image on the screen ?

(1) Keeping the positions of O and L unchanged, move S suitably closer to L.
(2) Keeping the positions of L and S unchanged, move O suitably farther away from L.
(3) Keeping the positions of O and S unchanged, move L suitably closer to S.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

2021 Q18

18. In the figure below, PQ and OS are light rays refracted from a lens L. Both light rays come from a point object situated on the left of L.



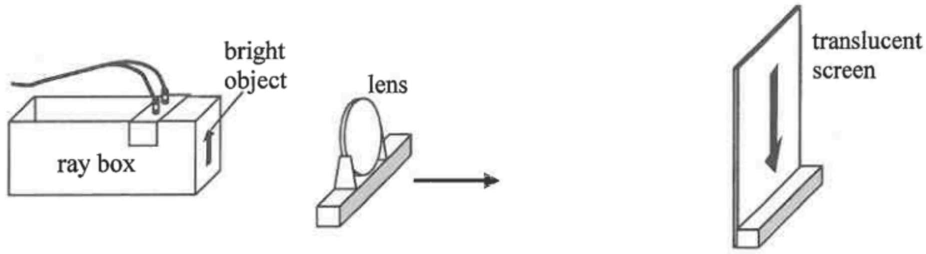
Which of the following deductions is/are correct ?

(1) The image of the object must be virtual.
(2) The object must lie along the straight line containing OS.
(3) L must be a concave lens.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

2022 Q21

21. In the set-up below, the separation between the bright object and the translucent screen is fixed. A lens is moved from the object towards the screen as shown.



The first sharp image formed is inverted as shown and its length is 9 cm. When the lens is moved further towards the screen, a second sharp image of length 1 cm is observed. Which of the following statements is/are correct ?

(1) The second image is erect.
(2) The length of the object is 3 cm.
(3) There are at most two positions of the lens that can give a sharp image on the screen when the lens is moved.

A. (1) only
B. (2) only
C. (2) and (3) only
D. (1), (2) and (3)

2023 Q20

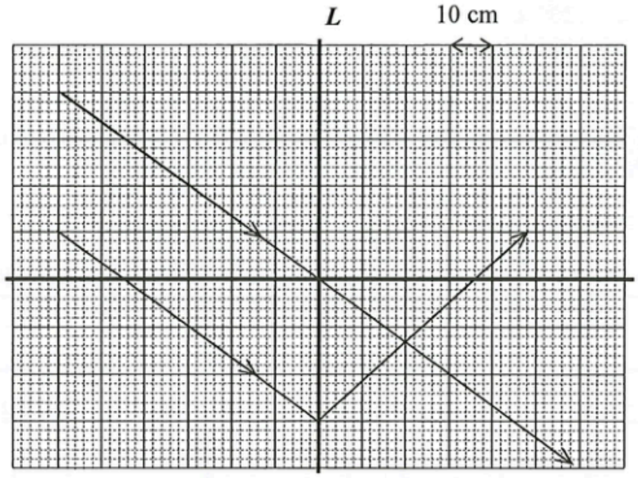
20. Virtual images of an object formed by a single lens are always

(1) erect.
(2) on the opposite side of the lens to the object.
(3) smaller than the object.

A. (1) only
B. (3) only
C. (2) and (3) only
D. (1), (2) and (3)

2023 Q21

21.

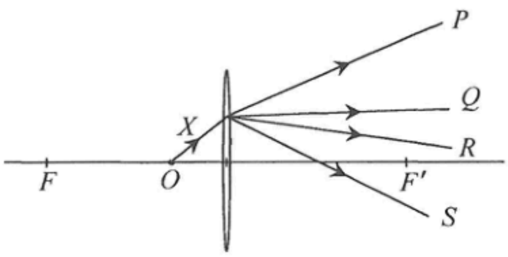


The ray diagram shows the refraction of two parallel light rays through a lens L. What is the type and the focal length of L ?

	type	focal length
A.	convex lens	20 cm
B.	convex lens	36 cm
C.	concave lens	20 cm
D.	concave lens	36 cm

2024 Q19

19. X is a light ray from an object O placed on the principal axis of a convex lens as shown. F and F' are the foci of the lens. Which light ray best represents the emergent ray ?



A. P
B. Q
C. R
D. S

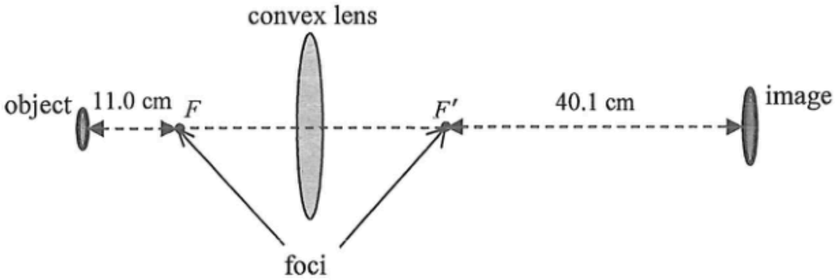
2024 Q20

*20. An object of height 10 cm placed at 20 cm in front of a concave lens forms an image of height 8 cm. Find the focal length of the concave lens.

A. 4.4 cm
B. 8.9 cm
C. 40 cm
D. 80 cm

2025 Q19

*19. An object is placed at a certain distance from a convex lens, the position of a screen is adjusted until a sharp image is formed.

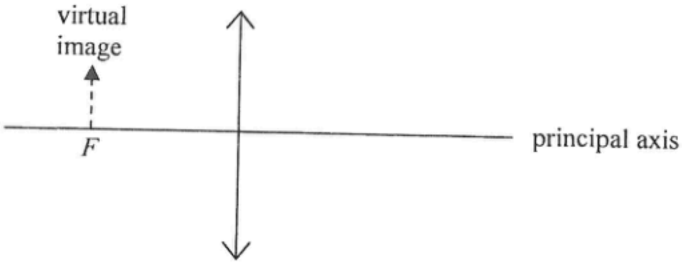


Given that the distances of the object and the image to the respective foci F and F' of the convex lens are 11.0 cm and 40.1 cm as shown, the focal length of the lens is

A. 8.6 cm.
B. 17.3 cm.
C. 21.0 cm.
D. 25.9 cm.

2026 Q16

16. The image of an object formed by a convex lens is virtual and is situated at the focus F as shown.

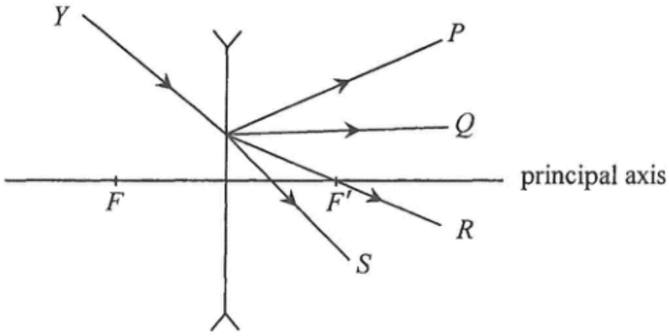


Which of the following statements is/are correct ?

(1) The object is at infinity.
(2) The object is larger than the image.
(3) The object and the image are on the same side of the lens.

A. (2) only
B. (3) only
C. (1) and (2) only
D. (1) and (3) only

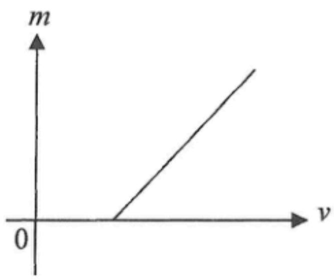
17. Light ray Y is incident on a concave lens as shown. F and F' are the foci of the lens. Which light ray best represents the emergent ray ?



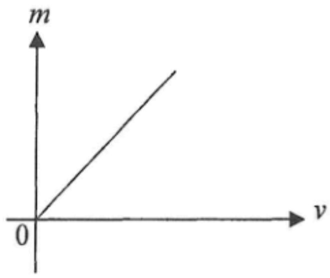
A. P
B. Q
C. R
D. S

*18. A convex lens is used to project sharp images of an object onto a screen at different object distances u. For each value of u, the corresponding image distance v and the linear magnification m are recorded. Which of the following best represents the graph of m against v ?

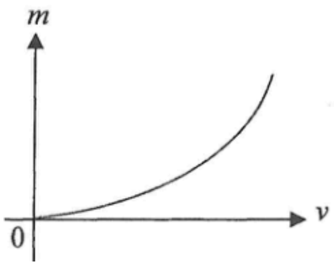
A.



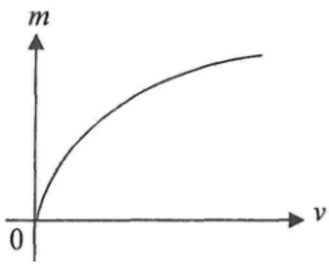
B.



C.



D.



*21.

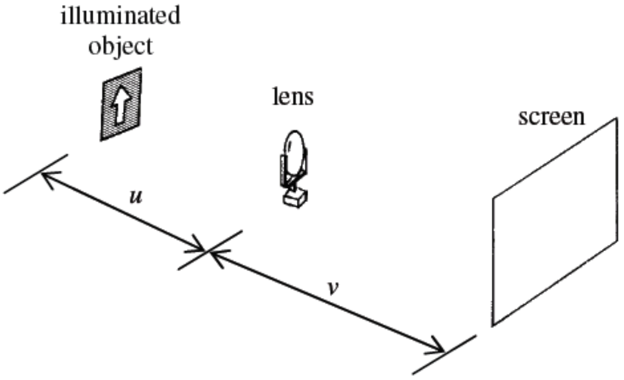


Figure (a)

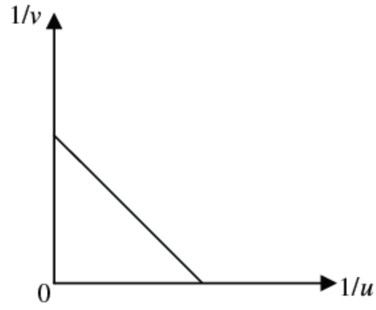
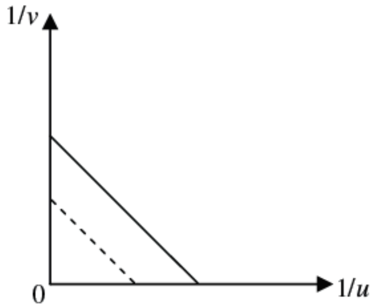


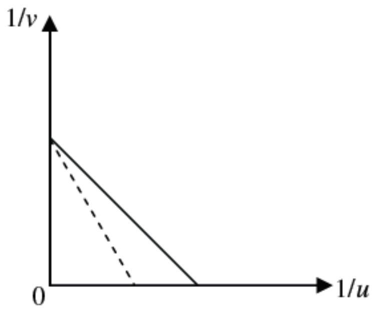
Figure (b)

A student uses the set-up in Figure (a) to study the relationship between the object distance u and the image distance v of a convex lens. A graph of $1/v$ against $1/u$ is plotted in Figure (b). If the lens is replaced by another convex lens of shorter focal length, which of the following graphs (in dotted lines) would be obtained ?

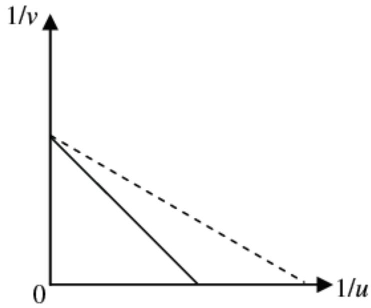
A.



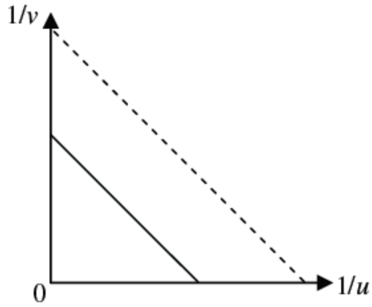
B.



C.

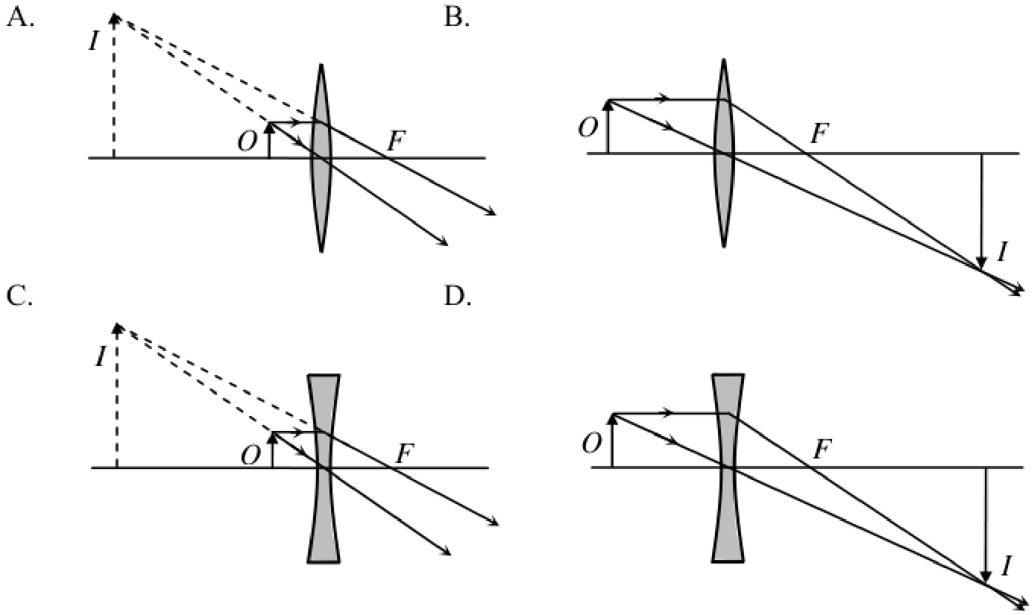


D.



16.

Cecilia uses a magnifying glass to read some small print. Which diagram shows how the image of tl print is formed ?



21. An object is placed at the focus of a concave lens of focal length 10 cm. What is the magnification of the image formed ?

A. 0.5
B. 1.0
C. 2.0
D. infinite